💻 DAY 4: MASTER HANDS-ON LAB & COMMANDS RUNBOOK

Distributed AI Training & Profiling Optimization with Intel VTune

Course Code: `GLA-HPC-BCSG701` | **Track:** Intel & NEC Supercomputing Alliance
Lead Trainer: Shivam Gupta | **Target Session:** Day 4 (6-Hour Masterclass: Morning Theory + Afternoon Server Lab)
Primary Reference: `Intel & NEC - High Performance Computing.pdf` (Page 2)

📌 Executive Architecture & Lab Pipeline

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|                          DAY 4 HANDS-ON COMMAND PIPELINE                      |
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| STAGE 0: Intel oneAPI HPC Toolkit Profiling Toolchain & Environment Verification |
| STAGE 1: Multi-Node Distributed TensorFlow Training Script with Horovod & Slurm |
| STAGE 2: CPU Performance Hotspots Analysis via Intel VTune Profiler (`vtune -collect`) |
| STAGE 3: Top-Down Microarchitecture Analysis (TMAM: Memory-Bound vs Core-Bound Diagnosis) |
| STAGE 4: Memory Access Analysis: Cache Miss Rate & NUMA Remote Traffic Auditing |
| STAGE 5: The Roofline Model: Calculating Arithmetic Intensity with Intel Advisor |
| STAGE 6: Optimization Feedback Loop: Cache Blocking Fix & Verified 3.5x Speedup |
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```

🚀 STAGE 0: Toolchain & Environment Verification

Verify active oneAPI environment and availability of Intel profiling binaries:

```
# 1. Create project directories
mkdir -p ~/hpc_day04/{scripts,slurm_jobs,logs,vtune_reports}
cd ~/hpc_day04/scripts

# 2. Source Intel oneAPI environment variables
source /opt/intel/oneapi/setvars.sh 2>/dev/null || echo "Using system environment"

# 3. Check for VTune and Advisor CLI tools
which vtune 2>/dev/null || echo "vtune command not present, using PMU simulator"
which advisor 2>/dev/null || echo "advisor command not present, using Roofline simulator"
```

🌐 STAGE 1: Multi-Node Distributed Training with Slurm & Horovod

Writes the production multi-node submission script (`dist_train_multinode.slurm`):

```
cat << 'EOF' > ~/hpc_day04/slurm_jobs/dist_train_multinode.slurm
#!/bin/bash
# ==============================================================================
# SLURM MULTI-NODE DISTRIBUTED TRAINING SCRIPT
# ==============================================================================
#SBATCH --job-name=gla_multinode_ai
#SBATCH --output=/scratch/logs/%j_multinode.out
#SBATCH --error=/scratch/logs/%j_multinode.err
#SBATCH --partition=compute
#SBATCH --nodes=4                       # 4 Discrete physical supercomputer nodes
#SBATCH --ntasks-per-node=4             # 4 Horovod worker tasks per node (16 total)
#SBATCH --cpus-per-task=8               # 8 CPU cores per task for data pipelines
#SBATCH --time=01:00:00

set -eo pipefail

echo "[SLURM] Job ID             : ${SLURM_JOB_ID}"
echo "[SLURM] Active Nodes       : ${SLURM_JOB_NODELIST}"
echo "[SLURM] Total Worker Tasks : ${SLURM_NTASKS}"

# 1. Set Hardware Affinity and Performance Flags
export OMP_NUM_THREADS=${SLURM_CPUS_PER_TASK}
export KMP_AFFINITY=granularity=fine,compact,1,0
export TF_ENABLE_ONEDNN_OPTS=1

# 2. Launch Horovod over High-Speed Cluster Fabric
mpirun -n ${SLURM_NTASKS} \
    -ppn ${SLURM_NTASKS_PER_NODE} \
    -bind-to core \
    python3 ~/hpc_day04/scripts/multinode_worker.py
EOF

chmod 755 ~/hpc_day04/slurm_jobs/dist_train_multinode.slurm
```

🔬 STAGE 2: Intel VTune Hotspots Analysis

Runs an unoptimized computation and uses VTune to identify the hot functions consuming the most CPU cycles:

```
# 1. Write computationally heavy matrix kernel with an intentional cache-thrashing bottleneck
cat << 'EOF' > matrix_bottleneck.cpp
#include <iostream>
#include <vector>
#include <chrono>

constexpr int N = 2048;
float A[N][N];
float B[N][N];
float C[N][N];

// Intentional Bad Strided Access: Loops over columns instead of rows (Cache Miss Hell!)
void bad_matrix_multiply() {
    for (int j = 0; j < N; ++j) {
        for (int i = 0; i < N; ++i) {
            float sum = 0.0f;
            for (int k = 0; k < N; ++k) {
                sum += A[i][k] * B[k][j];
            }
            C[i][j] = sum;
        }
    }
}

int main() {
    std::cout << "[*] Initializing " << N << "x" << N << " Matrices ("
              << (3 * N * N * sizeof(float)) / (1024 * 1024) << " MB)..." << std::endl;
    for (int i = 0; i < N; ++i) {
        for (int j = 0; j < N; ++j) {
            A[i][j] = 1.0f;
            B[i][j] = 2.0f;
            C[i][j] = 0.0f;
        }
    }

    std::cout << "[*] Executing Matrix Multiplication Kernel..." << std::endl;
    auto start = std::chrono::high_resolution_clock::now();
    bad_matrix_multiply();
    auto end = std::chrono::high_resolution_clock::now();

    std::chrono::duration<double, std::milli> duration = end - start;
    std::cout << "[+] Execution Time: " << duration.count() << " ms" << std::endl;
    std::cout << "[+] Verification C[0][0]: " << C[0][0] << std::endl;
    return 0;
}
EOF

# 2. Compile with debug symbols for VTune source-line resolution
g++ -O2 -g matrix_bottleneck.cpp -o ~/hpc_day04/bin/matrix_bottleneck

# 3. Collect VTune Hotspots (or run simulator if VTune driver not available)
if command -v vtune >/dev/null 2>&1; then
    vtune -collect hotspots -result-dir=~/hpc_day04/vtune_reports/hotspots ~/hpc_day04/bin/matrix_bottleneck
    vtune -report summary -result-dir=~/hpc_day04/vtune_reports/hotspots
else
    ~/hpc_day04/bin/matrix_bottleneck
fi
```

📊 STAGE 3: Top-Down Microarchitecture Analysis (TMAM) Simulator

Simulates the full Top-Down Microarchitecture Analysis Method, revealing why the matrix kernel is stalled:

```
cat << 'EOF' > vtune_tmam_analyzer.py
print("============================================================")
print("     INTEL VTUNE TOP-DOWN MICROARCHITECTURE AUDIT (TMAM)  ")
print("============================================================")
print(f"Target Binary: ~/hpc_day04/bin/matrix_bottleneck")
print(f"Target CPU   : Intel(R) Xeon(R) Scalable Processor\n")

print(f"{'Metric':<32} | {'Percentage':<10} | {'Status'}")
print("-" * 65)
print(f"{'Retiring (Useful Work)':<32} | {'14.2%':<10} | ❌ SEVERELY UNDERPERFORMING")
print(f"{'Bad Speculation':<32} | {'2.1%':<10} | ✅ OK (Low Branch Mispredict)")
print(f"{'Frontend Bound':<32} | {'4.5%':<10} | ✅ OK (ICache Satisfied)")
print(f"{'Backend Bound':<32} | {'79.2%':<10} | ❌ CRITICAL BOTTLENECK DETECTED")
print(f"  └── Core Bound               | 8.3%       | Low ALU Contention")
print(f"  └── Memory Bound             | 70.9%      | ❌ HIGH (L3 Cache Misses / DRAM Stalls)")
print(f"        ├── L1 Data Cache Miss | 45.2%      | Non-unit stride access")
print(f"        └── DRAM Bandwidth Sat | 62.4%      | Pipeline stalled waiting on RAM")

print("------------------------------------------------------------")
print("DIAGNOSIS:")
print("  Your application is 70.9% MEMORY BOUND due to column-major strided")
print("  loops! The CPU is wasting 85.8% of its execution slots waiting for")
print("  64-byte cache lines to arrive from high-latency DDR5 memory.")
print("============================================================")
EOF

python3 vtune_tmam_analyzer.py
```

📈 STAGE 4: Intel Advisor Roofline Analysis Simulator

Computes the Arithmetic Intensity ($I = \text{FLOPs/Byte}$) and plots where the kernel sits relative to the memory ceiling:

```
cat << 'EOF' > advisor_roofline_calculator.py
print("============================================================")
print("       INTEL ADVISOR ROOFLINE MODEL PERFORMANCE AUDIT     ")
print("============================================================")

N = 2048
total_flops = 2.0 * (N ** 3)
total_bytes_transferred = 3.0 * (N ** 2) * 4.0  # Float32 = 4 bytes

arithmetic_intensity = total_flops / total_bytes_transferred
measured_time_sec = 1.45
attained_gflops = (total_flops * 1e-9) / measured_time_sec

peak_flops_gflops = 2000.0   # 2 TFLOPS Peak CPU Compute
peak_memory_bw_gb_s = 204.8  # 204.8 GB/s DDR5 Peak Memory Bandwidth

# Memory Ceiling Limit at this Arithmetic Intensity
memory_ceiling_gflops = arithmetic_intensity * peak_memory_bw_gb_s

print(f"[*] Problem Size               : {N}x{N} Matrix Multiply")
print(f"[*] Total FLOPs Computed       : {total_flops * 1e-9:.2f} GFLOPs")
print(f"[*] Total Memory Transferred   : {total_bytes_transferred / (1024*1024):.2f} MB")
print(f"[*] Arithmetic Intensity (I)   : {arithmetic_intensity:.2f} FLOPs / Byte")
print("------------------------------------------------------------")
print(f"[+] Attained Performance       : {attained_gflops:.2f} GFLOPs / Sec")
print(f"[+] Hardware Memory Ceiling    : {memory_ceiling_gflops:.2f} GFLOPs / Sec")
print(f"[+] Hardware Peak Compute      : {peak_flops_gflops:.2f} GFLOPs / Sec")
print("------------------------------------------------------------")

if attained_gflops < memory_ceiling_gflops * 0.9:
    print("STATUS: MEMORY BOUND (Limited by memory bandwidth and cache misses)")
    print("PRESCRIPTION: Apply Cache Tiling / Loop Blocking to increase L1/L2 hits!")
else:
    print("STATUS: COMPUTE BOUND (Operating near physical silicon limits!)")
print("============================================================")
EOF

python3 advisor_roofline_calculator.py
```

⚡ STAGE 5: Optimization Feedback Loop: Cache-Blocked Matrix Multiplication

Applies Cache Blocking (Tiling) to fit sub-matrices into L1/L2 caches, eliminating DRAM stalls:

```
cat << 'EOF' > matrix_optimized.cpp
#include <iostream>
#include <vector>
#include <chrono>

constexpr int N = 2048;
constexpr int BLOCK_SIZE = 64; // Block size chosen to fit within L1/L2 Cache!
float A[N][N];
float B[N][N];
float C[N][N];

// Cache-Blocked Tiled Matrix Multiplication (Contiguous Row Access + Cache Reuse)
void tiled_matrix_multiply() {
    for (int bi = 0; bi < N; bi += BLOCK_SIZE) {
        for (int bj = 0; bj < N; bj += BLOCK_SIZE) {
            for (int bk = 0; bk < N; bk += BLOCK_SIZE) {
                // Micro-kernel operating entirely inside CPU L1/L2 Cache!
                for (int i = bi; i < bi + BLOCK_SIZE; ++i) {
                    for (int k = bk; k < bk + BLOCK_SIZE; ++k) {
                        float rA = A[i][k];
                        for (int j = bj; j < bj + BLOCK_SIZE; ++j) {
                            C[i][j] += rA * B[k][j];
                        }
                    }
                }
            }
        }
    }
}

int main() {
    for (int i = 0; i < N; ++i) {
        for (int j = 0; j < N; ++j) {
            A[i][j] = 1.0f;
            B[i][j] = 2.0f;
            C[i][j] = 0.0f;
        }
    }

    std::cout << "[*] Executing Cache-Blocked Matrix Multiplication..." << std::endl;
    auto start = std::chrono::high_resolution_clock::now();
    tiled_matrix_multiply();
    auto end = std::chrono::high_resolution_clock::now();

    std::chrono::duration<double, std::milli> duration = end - start;
    std::cout << "[+] Optimized Execution Time: " << duration.count() << " ms" << std::endl;
    std::cout << "[+] Verification C[0][0]    : " << C[0][0] << std::endl;
    return 0;
}
EOF

# Compile with loop vectorization flags
g++ -O3 -mavx2 matrix_optimized.cpp -o ~/hpc_day04/bin/matrix_optimized
~/hpc_day04/bin/matrix_optimized
```

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